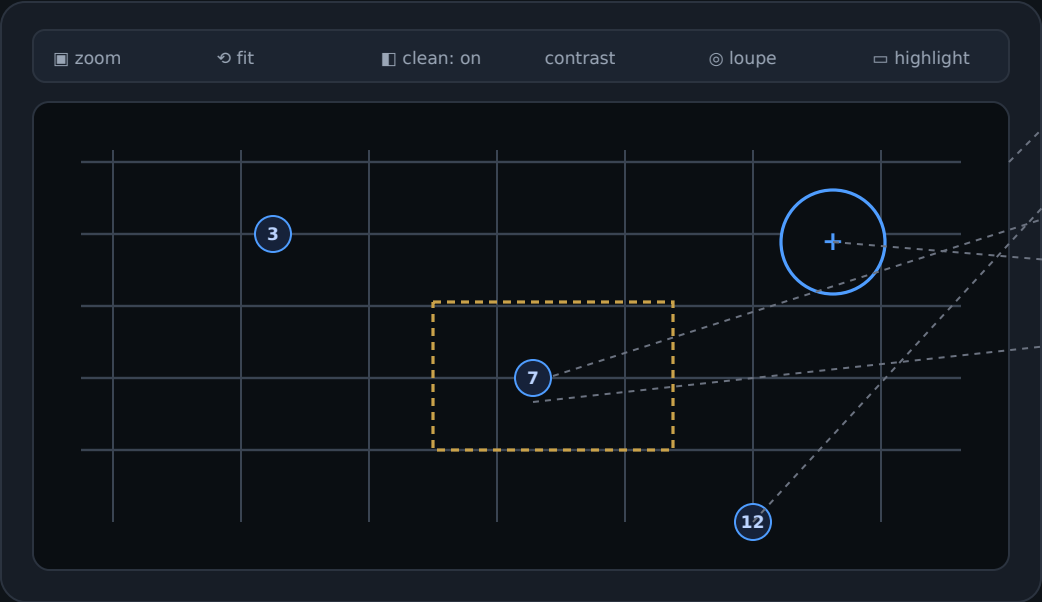


Readable, interactive schematics — options & how it works

It's the SAME drawing — enhanced for legibility (contrast/sharpen/de-speckle), deep-zoomable, with hover/highlight. No content invented; everything cited to its page.

THE VIEWER (marked up)



1 deep zoom / pan to any portion

2 callout numbers hover → highlight + open the part

3 spotlight / box-highlight a section

4 magnifier loupe on hover

5 'clean' toggle + contrast = legibility

Example (real page): see the before/after image shared with this.

LEGIBILITY PIPELINE (offline, grounded)

1 · Render page hi-DPI (PyMuPDF)

crisp raster of the real page

2 · Grayscale + auto-contrast

stretch faded scans

3 · Sharpen (unsharp) + de-speckle

crisper lines, less scan noise

4 · Optional: deskew · binarize · upscale

straighten · black-on-white · enlarge low-res

5 · Tile pyramid → deep-zoom viewer

OpenSeadragon: smooth zoom/pan + overlays

OPTIONS (grounded → richer)

A · Legibility cleanup

Contrast / sharpen / de-speckle / deskew + a 'clean' toggle and contrast slider. Proven on your scan (shared).

MINIMUM · NOW

low effort · it's the real drawing, enhanced

B · Deep zoom + pan

Tile the page; OpenSeadragon for smooth zoom to any portion, fully offline.

RESIZE/ZOOM

low-med effort

C · Hover loupe + box-highlight / spotlight

Magnifier that follows the cursor; drag to highlight a region or dim everything outside it.

HOVER/HIGHLIGHT

med effort

D · Callout hotspots → parts

Hover/click a FIG callout number to highlight it and open the part (NSN/part#). Ties to the parts index; needs OCR boxes.

SMART

med effort · OCR-gated

E · Vectorize the line art

Trace clean line drawings to SVG → infinite-zoom crispness + per-line hover/recolor. Best for clean schematics.

CRISPEST

high effort

GROUNDING + RECOMMENDATION

Enhancement (contrast, sharpen, de-speckle, deskew, interpolated upscale) is legitimate — it's the same drawing, more legible, cited to its page. Avoid AI super-resolution / inpainting that fabricates strokes (it can invent detail). Recommended bundle: A (cleanup) + B (deep zoom) + C (loupe/highlight) now; D (callout→part) as OCR completes; E (vectorize) optional for the cleanest line schematics.